BEYOND THE BOHR–KRONECKER TORUS BARRIER: SEMIPRIME PHASE-LOCKING, QUANTIZED HADAMARD–KREIN FLUX DEFECTS $\pi\,\Theta(d-d_*)$, AND THE EXACT CHLADNI–WRONSKIAN WALL IDENTITY FOR RIEMANN'S Ξ-FUNCTION

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October 25, 2026

Abstract. We establish three closure mechanisms for the parity-reduced transverse Wronskian $\widetilde{W}(d,t)=\frac{1}{2d}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's Ξ-function across $0<d\le\frac12$ beyond the Bohr–Kronecker torus barrier. First, the Bohr–Kronecker recurrence horizon for the $\pi(N(t))-\pi(X)$ active primes below $N(t)=\lfloor\sqrt{t/2\pi}\rfloor$ grows as $\exp(C\frac{\sqrt{t}}{\ln t}(1+o(1)))\gg t$, while in $\zeta(s)E_X(s)=\exp(\mathcal{P}_{>X}(s))$ ($\mathcal{P}_{>X}(s)=U+\mathrm{i}V$), semiprime phase-locking enforces $|1+\mathcal{P}_{>X}+\frac12\mathcal{P}_{>X}^2|^2=\frac14((1+U)^2+1-V^2)^2+V^2(1+U)^2\ge\frac14$ on $V=0$. Second, the mirror Hadamard kernel of any off-critical pair $z_\pm=\gamma_*\mp\mathrm{i}d_*$ integrates to $\pi\,\Theta(d-d_*)$, creating a $-\pi$ flux defect on $0<d<d_*$ via arcsine cancellation $\mp\arcsin(d/d_*)$ and forcing a Sylvester jump $\nu_-\ge1$ in the de Branges Gram matrix. Third, along every Chladni wall $\Gamma_k:\ I(d,\tau_k(d))\equiv0$, Cauchy–Riemann differentiation yields $\widetilde{W}(d,\tau_k(d))\equiv R(d,\tau_k(d))\,\partial_tI(d,\tau_k(d))/d$, with Jacobi triple-product zero-freeness in $|\mathrm{Im}(u)|<\frac{\pi}{8}$.

1. Introduction and Main Results

A central obstruction in the classical theory of Dirichlet series is the decoupling of prime phases on the infinite-dimensional torus $\mathbb{T}^\infty$, initiated by Bohr and Kronecker [1, 10] and quantified in Voronin's universality theorem [27, 26]. When analyzing the transverse Wronskian $W(d,t):=\frac12\partial_d|\Xi(t-\mathrm{i}d)|^2=R(d,t)\,\partial_tI(d,t)-I(d,t)\,\partial_tR(d,t)$ of Riemann's completed zeta function $\Xi(z):=\xi(\frac12+\mathrm{i}z)=R(d,t)+\mathrm{i}\,I(d,t)$ ($z=t-\mathrm{i}d$, $s=\frac12+d+\mathrm{i}t$) [23, 25, 6, 11] across the critical strip $0<d\le\frac12$, any purely additive treatment of the truncated Dirichlet series $\sum_{n\le N(t)}b_nn^{-s}$ up to the Riemann–Siegel saddle horizon $N(t):=\lfloor\sqrt{t/2\pi}\rfloor$ [24, 5] encounters three fundamental questions:

(1) *The Bohr–Kronecker anti-phase alignment paradox.* If the prime phases $\phi_p(t)=-t\ln p \pmod{2\pi}$ are treated as independent free variables on the torus $\mathbb{T}^{\pi(N(t))-\pi(X)}$, the triangle inequality permits an artificial worst-case state in which all active primes align near anti-phase $\phi_p\approx\pi$, even though the available spectral height t is exponentially smaller than the Kronecker recurrence time of the torus.

(2) *Topological flux quantization versus local Krein-space sign inversions.* In Hadamard's product factorization [22, 3] and de Branges Hilbert spaces of entire functions [4, 2], how does a hypothetical off-critical conjugate pair $z_\pm=\gamma_*\mp\mathrm{i}d_*$ ($d_*>0$) alter the global longitudinal Poisson flux and the Sylvester signature of the reproducing-kernel Gram matrix on horizontal lines $0<d<d_*$?

(3) *Bypassing 2D Dirichlet triangle inequalities via 1D nodal geometry.* In the Mellin–Jacobi integral representation $\Xi(z)=8\int_0^\infty\Phi(u)\cos(2zu)\,\mathrm{d}u$ [9, 28, 7], the generating theta series is a

Date: October 25, 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 11M26; Secondary 30D15, 46C20.

Key words and phrases. Riemann Ξ-function, transverse Wronskian, Bohr–Kronecker recurrence, semiprime phase-locking, Hadamard–Krein flux defect, de Branges reproducing kernel, Sylvester inertia index, Chladni nodal walls, Jacobi triple product.

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single holomorphic function in $|\mathrm{Im}(u)| < \frac{\pi}{8}$. Can the two-dimensional positivity of $\widetilde{W}(d,t) := W(d,t)/d$ inside each harmonic Chladni nodal chamber Ω_k be reduced to a one-dimensional holomorphic differential identity along the chamber boundary walls $\Gamma_k : I(d, \tau_k(d)) \equiv 0$?

Building upon the Wigner phase-space saddle coalescence, parity-reduced Wronskian, harmonic Chladni chamber, de Branges reproducing kernel, universal mother-wavelet, and exponential prime-gauge frameworks established in Papers I–IX of this series [12, 13, 14, 15, 16, 17, 18, 19, 20], the present work (Paper X) resolves these three questions through three exact analytical theorems. Our main results are consolidated as follows.

Theorem 1.1 (Semiprime Phase-Locking, Quantized Hadamard–Krein Flux Defect, and Chladni–Wronskian Wall Closure)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ and $z = t - \mathrm{i}d$ with $d > 0$ and $t > 0$, let $N(t) := \lfloor \sqrt{t/2\pi} \rfloor$, and let $\widetilde{W}(d,t) := \frac{1}{2d}\partial_d |\Xi(t - \mathrm{i}d)|^2$. Then the following three structural laws hold:*

(i) ***Exponential Bohr–Kronecker Recurrence Barrier and Semiprime Quadratic Restoration.*** *For any sieve cutoff $X = \lfloor (t/2\pi)^\theta \rfloor$ with $\theta \in [\frac{1}{6}, \frac{1}{4})$, the $M(t;X) := \pi(N(t)) - \pi(X)$ active prime oscillators in $(X, N(t)]$ possess $\mathbb{Q}$-linearly independent frequencies $\{\frac{\ln p}{2\pi}\}_{X<p\le N(t)}$ whose ergodic (Kac–Weyl) mean Bohr–Kronecker recurrence horizon for simultaneous ε-alignment ($\varepsilon \in (0,\pi)$) satisfies $T_{\mathrm{Kronecker}}(t;X,\varepsilon) = (\frac{\pi}{\varepsilon})^{M(t;X)} = \exp(\ln(\frac{\pi}{\varepsilon})\frac{\sqrt{2t/\pi}}{\ln(t/2\pi)}(1 + o(1))) \gg t$. Furthermore, in the exponential prime-gauge series $\zeta(s)E_X(s) = \exp(\mathcal{P}_{>X}(s)) = \sum_{k=0}^{\infty} \frac{1}{k!}\mathcal{P}_{>X}(s)^k$ with $\mathcal{P}_{>X}(s) = U(d,t) + \mathrm{i}V(d,t)$, multiplicative phase-locking $\phi_{pq}(t) = \phi_p(t) + \phi_q(t)$ between the linear prime shell ($k = 1$) and the quadratic semiprime shell ($k = 2$) enforces the exact algebraic sum-of-squares identity*

$$\Big|1 + \mathcal{P}_{>X}(s) + \frac{1}{2}\mathcal{P}_{>X}(s)^2\Big|^2 \equiv \frac{1}{4}\Big((1+U)^2 + 1 - V^2\Big)^2 + V^2(1+U)^2, \tag{1.1}$$

which on the real anti-phase axis $V = 0$ satisfies $|1 + U + \frac{1}{2}U^2|^2 \ge \frac{1}{4} > 0$ for all $U \in \mathbb{R}$, attaining its unique real-axis global minimum $\frac{1}{4}$ at $(U,V) = (-1,0)$.

(ii) ***Quantized Hadamard–Krein Flux-Defect Law $\pi\,\Theta(d - d_*)$ and Sylvester Inertia Jump.*** *For any off-critical conjugate zero pair $z_\pm = \gamma_* \mp \mathrm{i}d_*$ ($d_* > 0$) and relative coordinate $u = t - \gamma_*$, the mirror-paired Hadamard–Poisson kernel $K(d, d_*, u) := \frac{d(d^2 - d_*^2 + u^2)}{((d-d_*)^2+u^2)((d+d_*)^2+u^2)}$ integrates across $u \in (-\infty, \infty)$ to the quantized step function*

$$\mathcal{F}(d,d_*) := \int_{-\infty}^{\infty} K(d,d_*,u)\,\mathrm{d}u \equiv \frac{\pi}{2}\big(\mathrm{sgn}(d-d_*) + 1\big) = \pi\,\Theta(d-d_*) = \begin{cases} 0 & \text{if } 0 < d < d_*, \\ \frac{\pi}{2} & \text{if } d = d_*, \\ \pi & \text{if } d > d_*. \end{cases} \tag{1.2}$$

For every $0 < d < d_*$, the negative Krein lobe on $|u| < u_0(d,d_*) := \sqrt{d_*^2 - d^2}$ and the positive tails on $|u| > u_0(d,d_*)$ satisfy the exact arcsine cancellation law $\mathcal{F}_\mp(d,d_*) = \mp \arcsin(d/d_*)$, creating a $-\pi$ topological flux defect relative to a critical-line zero ($d_\rho = 0$, flux $+\pi$) and forcing a Sylvester negative inertia jump $\nu_-(\mathbf{G}_\alpha(d)) \ge 1$ in the de Branges reproducing-kernel Gram matrix $[\mathbf{G}_\alpha(d)]_{j,k} = K_\alpha(t_j - \mathrm{i}d, t_k - \mathrm{i}d)$, whereas in the u-domain every symplectic mode ellipse satisfies $|T_n(d,t)|^2 - |T_n(0,t)|^2 = a_n^2 \sinh^2(d\omega_n) > 0$.

(iii) ***The Exact Chladni–Wronskian Wall Identity and Jacobi Triple-Product Zero-Freeness.*** *Along any smooth imaginary nodal chamber wall $\Gamma_k : I(d, \tau_k(d)) \equiv 0$ departing orthogonally from a critical-line extremum $\tau_k(0) \in (\gamma_k, \gamma_{k+1})$ ($\Xi'(\tau_k(0)) = 0$, $\tau_k'(0) = 0$), implicit Cauchy–Riemann differentiation $\tau_k'(d) = -\partial_d I/\partial_t I = \partial_t R/\partial_d R$ yields the exact one-dimensional wall identities*

$$\widetilde{W}\big(d,\tau_k(d)\big) \equiv \frac{R(d,\tau_k(d))\,\partial_t I(d,\tau_k(d))}{d}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big|\Xi(\tau_k(d) - \mathrm{i}d)\Big|^2 \equiv 2d\,\widetilde{W}\big(d,\tau_k(d)\big)\Big(1 + \tau_k'(d)^2\Big), \tag{1.3}$$

while the Jacobi triple product $\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u}) = \prod_{n=1}^{\infty}(1-\mathrm{e}^{-2\pi n\mathrm{e}^{4u}})(1+\mathrm{e}^{-\pi(2n-1)\mathrm{e}^{4u}})^2 \neq 0$ *is strictly zero-free throughout the holomorphic saddle strip* $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned}
\min_{U\in\mathbb{R}}\left|1+U+\tfrac{1}{2}U^2\right|^2 &= \frac{1}{4} = 0.250000\ldots, & \log_{10} T_{\mathrm{Kronecker}}(10^8) &= +327.52\ \mathrm{dex},\\
\mathcal{F}_-(0.12, 0.20) &= -\arcsin(0.6) = -0.643501\ldots, & \Delta\mathcal{F}(0<d<d_*) &= -\pi = -3.141592\ldots,\\
\tau_1(0) &= 15.58570859\ldots, & \widetilde{W}(0.50, \tau_1(0.50)) &= +3.947735\times 10^{-7}.
\end{aligned} \tag{1.4}$$

1.1. **Proof overview.** The proof of Theorem 1.1 combines three quantitative ingredients developed in Sections 2–4:

(1) *Diophantine torus volume bounds and multiplicative quadratic phase-locking (Section 2).* Using the $\mathbb{Q}$-linear independence of $\{\ln p\}$ and Weyl's equidistribution on $\mathbb{T}^{M(t;X)}$, we establish the exponential recurrence lower bound $T_{\mathrm{Kronecker}}(t;X,\varepsilon) \geq (\pi/\varepsilon)^{\pi(N(t))-\pi(X)} \gg t$. Expanding $\zeta(s)E_X(s) = \exp(\mathcal{P}_{>X}(s))$ up to its quadratic semiprime shell $k=2$, we prove that the locked phase relation $\phi_{pq}(t) = \phi_p(t)+\phi_q(t)$ converts destructive real prime excursions $U \approx -1$ into constructive semiprime restoration $+\frac{1}{2}U^2 \approx +\frac{1}{2}$ via the exact algebraic completion of squares (1.1).

(2) *Contour integration of the mirror Hadamard kernel and Pontryagin–Krein signature analysis (Section 3).* Decomposing the mirror-paired Hadamard kernel $K(d,d_*,u)$ into partial fractions at $d-d_*$ and $d+d_*$, we evaluate the improper integral (1.2) in closed form and establish the trigonometric identity $\arctan(\frac{\sqrt{d_*^2-d^2}}{d_*-d}) - \arctan(\frac{\sqrt{d_*^2-d^2}}{d_*+d}) \equiv \arcsin(d/d_*)$ for the Krein inversion lobe $|u| < \sqrt{d_*^2-d^2}$. Coupling this with the de Branges reproducing-kernel identity $K_\alpha(z,z) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$ from Paper IV [15], we prove that any off-critical zero pair forces a Pontryagin–Krein signature defect $\nu_-(\mathbf{G}_\alpha(d)) \geq 1$ [8].

(3) *Implicit Cauchy–Riemann geometry on Chladni walls and Jacobi triple-product non-vanishing (Section 4).* Differentiating the nodal level set $I(d,\tau_k(d)) = 0$ implicitly and applying the Cauchy–Riemann equations $\partial_d R = +\partial_t I$ and $\partial_d I = -\partial_t R$ in $z = t - \mathrm{i}d$, we eliminate the imaginary cross-term $I\,\partial_d I \equiv 0$ along Γ_k to obtain (1.3). Finally, we invoke Jacobi's triple product for $\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u})$ to prove that $|\mathrm{e}^{-\pi\mathrm{e}^{4u}}| = \exp(-\pi\mathrm{e}^{4\xi}\cos 4\eta) < 1$ strictly throughout $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with $d = \sigma - \frac{1}{2} \in [0, \frac{1}{2}]$ and $t > 0$, and $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$. Riemann's completed zeta function is $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ and $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = R(d,t) + \mathrm{i}\,I(d,t)$. Because $z = t - \mathrm{i}d$ has imaginary part $y = -d$, the Cauchy–Riemann equations for the holomorphic function $\Xi(z)$ take the canonical form

$$\partial_d R = +\partial_t I, \qquad \partial_d I = -\partial_t R, \qquad |\Xi'(t-\mathrm{i}d)|^2 = (\partial_t R)^2 + (\partial_t I)^2 = (\partial_d I)^2 + (\partial_d R)^2. \tag{1.5}$$

The transverse Wronskian is $W(d,t) := \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = R\,\partial_d R + I\,\partial_d I = R\,\partial_t I - I\,\partial_t R$, and the parity-reduced Wronskian is $\widetilde{W}(d,t) := W(d,t)/d$. In the logarithmic u-domain, $\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u$, where $\Phi(u) := \sum_{m=1}^{\infty}(2\pi^2m^4\mathrm{e}^{9u} - 3\pi m^2\mathrm{e}^{5u})\exp(-\pi m^2\mathrm{e}^{4u})$. All high-precision numerical values are computed at 60-decimal working precision (`mpmath.mp.dps = 60`) and certified via 35-decimal directed-rounding interval arithmetic (`mpmath.iv.dps = 35`) in `scripts/verify_paper10_spectral_resonance_chladni_closure.py`.

2. The Bohr–Kronecker Recurrence Paradox and Multiplicative Semiprime Phase-Locking

In classical additive estimates for Dirichlet polynomials (see Bohr [1], Montgomery and Vaughan [21], and Voronin [27]), the prime phases $\phi_p(t) := -t \ln p \pmod{2\pi}$ are embedded into an abstract torus $\mathbb{T}^M$, where Kronecker's theorem [10] asserts that the trajectory $\mathbf{\Phi}(t) = (\phi_{p_1}(t), \dots, \phi_{p_M}(t))$ is dense as $t \to \infty$. However, in the Riemann–Siegel formula [24, 5], the number of active oscillators $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ is not fixed: the dimension $M(t; X) = \pi(N(t)) - \pi(X)$ of the torus grows simultaneously with t.

Theorem 2.1 (Exponential Bohr–Kronecker Recurrence Horizon vs. Saddle Horizon Growth)**.** *Let $t \geq 10^3$, let $N(t) := \lfloor \sqrt{t/2\pi} \rfloor$, and let $X = \lfloor (t/2\pi)^\theta \rfloor$ with $\theta \in [0, \frac{1}{4})$. Let $M(t; X) := \pi(N(t)) - \pi(X)$ denote the number of active primes $p \in (X, N(t)]$. For any angular tolerance $\varepsilon \in (0, \pi)$, let $\mathcal{A}_\varepsilon \subset \mathbb{T}^{M(t;X)}$ be the target anti-phase hypercube*

$$\mathcal{A}_\varepsilon := \Big\{ \theta = (\theta_p)_{X < p \leq N(t)} \in \mathbb{T}^{M(t;X)} : |\theta_p - \pi| \leq \varepsilon \ \text{ for all } X < p \leq N(t) \Big\}. \tag{2.1}$$

Then the normalized Haar measure of $\mathcal{A}_\varepsilon$ on $\mathbb{T}^{M(t;X)} = (\mathbb{R}/2\pi\mathbb{Z})^{M(t;X)}$ is $\mu_{\mathrm{Haar}}(\mathcal{A}_\varepsilon) = (\frac{\varepsilon}{\pi})^{M(t;X)}$, and by Weyl's equidistribution theorem and Dirichlet's box principle on the $\mathbb{Q}$-linearly independent frequencies $\{\frac{\ln p}{2\pi}\}_{X < p \leq N(t)}$, the ergodic (Kac–Weyl) mean Bohr–Kronecker recurrence horizon $T_{\mathrm{Kronecker}}(t; X, \varepsilon) := \mu_{\mathrm{Haar}}(\mathcal{A}_\varepsilon)^{-1}$ required to visit $\mathcal{A}_\varepsilon$ satisfies

$$T_{\mathrm{Kronecker}}(t; X, \varepsilon) = \left(\frac{\pi}{\varepsilon}\right)^{\pi(N(t)) - \pi(X)} = \exp\left(\ln\left(\frac{\pi}{\varepsilon}\right) \frac{\sqrt{2t/\pi}}{\ln(t/2\pi)} (1 + o(1)) \right) \gg t \qquad (t \to \infty). \tag{2.2}$$

Proof. By the fundamental theorem of arithmetic, the logarithms $\{\ln p\}_{X < p \leq N(t)}$ are linearly independent over $\mathbb{Q}$. Consequently, by Weyl's theorem on uniform distribution modulo 2π, the asymptotic fraction of spectral height $t \in [0, T]$ for which the phase vector $\mathbf{\Phi}(t) = (-t \ln p \bmod 2\pi)_{X < p \leq N(t)}$ lies inside the hypercube $\mathcal{A}_\varepsilon$ of side length 2ε is given exactly by the product of the one-dimensional angular ratios:

$$\lim_{T \to \infty} \frac{1}{T} \operatorname{meas}\{ t \in [0, T] : \mathbf{\Phi}(t) \in \mathcal{A}_\varepsilon \} = \prod_{X < p \leq N(t)} \frac{2\varepsilon}{2\pi} = \left(\frac{\varepsilon}{\pi}\right)^{M(t;X)}.$$

Similarly, partitioning $\mathbb{T}^{M(t;X)}$ into $\lceil \pi/\varepsilon \rceil^{M(t;X)}$ congruent boxes of radius ε via Dirichlet's box principle shows that the mean recurrence time $T_{\mathrm{Kronecker}}(t; X, \varepsilon)$ scales as $(\pi/\varepsilon)^{M(t;X)}$. By the prime number theorem, $\pi(N(t)) = \frac{N(t)}{\ln N(t)}(1 + o(1)) = \frac{2\sqrt{t/2\pi}}{\ln(t/2\pi)}(1 + o(1))$ while $\pi(X) = \mathcal{O}(t^\theta / \ln t) = o(\pi(N(t)))$ since $\theta < \frac{1}{4}$. Taking logarithms yields $\ln T_{\mathrm{Kronecker}}(t; X, \varepsilon) = M(t; X) \ln(\pi/\varepsilon) = \ln(\pi/\varepsilon) \frac{\sqrt{2t/\pi}}{\ln(t/2\pi)}(1 + o(1))$, which grows exponentially faster than $\ln t$ as $t \to \infty$. □

Beyond this kinematic timescale separation, the exponential prime-gauge factor $E_X(s) = \exp(-\sum_{n=2}^{\lfloor X \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s})$ introduced in Paper VI [17] provides a deterministic algebraic restoration mechanism whenever $\theta \in [\frac{1}{6}, \frac{1}{4})$. In that regime, $X^2 < N(t) \leq X^3$, so the active Riemann–Siegel main sum $1 \leq n \leq N(t)$ contains not only the linear prime-power shell ($k = 1$: $p^m \in (X, N(t)]$), but also the quadratic semiprime shell ($k = 2$: $n = pq \in (X^2, N(t)]$ with $X < p \leq q$). Because $n^{-s} = (pq)^{-s} = p^{-s} q^{-s}$, the semiprime phases are multiplicatively locked to the prime phases via $\phi_{pq}(t) \equiv \phi_p(t) + \phi_q(t)$.

Theorem 2.2 (Semiprime Phase-Locking and Exact Quadratic Sum-of-Squares Restoration)**.** *Let $X \geq 2$ and let $\mathcal{P}_{>X}(s) := \sum_{n > \lfloor X \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s} = U(d, t) + \mathrm{i} V(d, t)$ be the residual prime-power Dirichlet series in the exact generating identity $\zeta(s) E_X(s) = \exp(\mathcal{P}_{>X}(s)) = \sum_{k=0}^{\infty} \frac{1}{k!} \mathcal{P}_{>X}(s)^k$. Then:*

(i) ***Cubic Semiprime Window on*** $(X,(\lfloor X\rfloor+1)^3)$**.** *For every integer* $1\le N<(\lfloor X\rfloor+1)^3$*, all shells* $k\ge 3$ *in* $\exp(\mathcal{P}_{>X}(s))$ *vanish identically on* $[1,N]$*, and the sieved Dirichlet coefficients* $b_N^{\mathcal{E}}(X)$ *of* $\zeta(s)E_X(s)=\sum_{N=1}^{\infty}b_N^{\mathcal{E}}(X)N^{-s}$ *are given exactly by the sum of the* $k\in\{0,1,2\}$ *shells:*

$$b_N^{\mathcal{E}}(X)=\delta_{N,1}+\mathbf{1}_{N>\lfloor X\rfloor}\frac{\Lambda(N)}{\ln N}+\frac{1}{2}\sum_{\substack{n_1n_2=N\\ n_1,n_2>\lfloor X\rfloor}}\frac{\Lambda(n_1)}{\ln n_1}\frac{\Lambda(n_2)}{\ln n_2}\qquad(1\le N<(\lfloor X\rfloor+1)^3).\tag{2.3}$$

In particular, $b_1^{\mathcal{E}}(X)=1$*,* $b_N^{\mathcal{E}}(X)=0$ *for all* $2\le N\le\lfloor X\rfloor$*,* $b_p^{\mathcal{E}}(X)=1$ *for every prime* $p>\lfloor X\rfloor$*, and* $b_{pq}^{\mathcal{E}}(X)=1$ *for every semiprime* $N=pq<(\lfloor X\rfloor+1)^3$ *with* $\lfloor X\rfloor<p\le q$*.*

(ii) ***Exact Quadratic Sum-of-Squares Identity.*** *The quadratic semiprime truncation* $F_2(U,V):=1+\mathcal{P}_{>X}(s)+\frac{1}{2}\mathcal{P}_{>X}(s)^2$ *satisfies the exact global algebraic identity*

$$\boxed{\left|1+\mathcal{P}_{>X}(s)+\frac{1}{2}\mathcal{P}_{>X}(s)^2\right|^2\equiv\frac{1}{4}\Big((1+U)^2+1-V^2\Big)^2+V^2(1+U)^2.}\tag{2.4}$$

(iii) ***Anti-Phase Restoration Spring.*** *The real part of the quadratic truncation satisfies* $\operatorname{Re}F_2(U,V)=\frac{1}{2}((1+U)^2+1-V^2)$*. On the real anti-phase axis* $V=0$ *(and more generally whenever* $|V|<1$*),* $|F_2(U,V)|^2$ *is strictly positive for all* $U\in\mathbb{R}$*:*

$$\left|1+U+\frac{1}{2}U^2\right|^2\equiv\frac{1}{4}\Big((1+U)^2+1\Big)^2\ge\frac{1}{4}>0,\qquad\operatorname{Re}\Big(1+U+\frac{1}{2}U^2\Big)=\frac{1}{2}\Big((1+U)^2+1\Big)\ge\frac{1}{2}>0,\tag{2.5}$$

attaining its unique real-axis global minimum at the anti-phase point $(U,V)=(-1,0)$*, where the linear prime trough* $U=-1$ *automatically excites the in-phase semiprime restoration* $+\frac{1}{2}U^2=+\frac{1}{2}$*.*

Proof. (i) Every term in $\mathcal{P}_{>X}(s)$ has index $n\ge\lfloor X\rfloor+1$. Hence any k-fold product in $\frac{1}{k!}\mathcal{P}_{>X}(s)^k$ with $k\ge3$ has integer index $N=n_1\cdots n_k\ge(\lfloor X\rfloor+1)^3$. Thus only $k\in\{0,1,2\}$ contribute to $1\le N<(\lfloor X\rfloor+1)^3$. For two distinct primes $p,q>\lfloor X\rfloor$, the cross-term in $\frac{1}{2!}\mathcal{P}_{>X}(s)^2$ is $\frac{1}{2}(p^{-s}q^{-s}+q^{-s}p^{-s})=(pq)^{-s}$, giving $b_{pq}^{\mathcal{E}}(X)=1$. For $p=q>\lfloor X\rfloor$, the linear $k=1$ prime-square term $\frac{1}{2}p^{-2s}$ and the quadratic $k=2$ term $\frac{1}{2!}(p^{-s})^2=\frac{1}{2}p^{-2s}$ sum to $b_{p^2}^{\mathcal{E}}(X)=\frac{1}{2}+\frac{1}{2}=1$.

(ii)–(iii) Substituting $\mathcal{P}_{>X}(s)=U+\mathrm{i}V$ into $F_2(U,V)=1+\mathcal{P}_{>X}+\frac{1}{2}\mathcal{P}_{>X}^2$ and separating real and imaginary parts gives

$$F_2(U,V)=\left(1+U+\frac{1}{2}U^2-\frac{1}{2}V^2\right)+\mathrm{i}\,V(1+U)=\frac{1}{2}\Big((1+U)^2+1-V^2\Big)+\mathrm{i}\,V(1+U).$$

Taking the squared modulus $|\operatorname{Re}F_2|^2+|\operatorname{Im}F_2|^2$ immediately yields (2.4). Setting $V=0$ gives (2.5), whose unique stationary minimum occurs at $1+U=0\iff U=-1$. □

Table 1 summarizes the exponential divergence of the Bohr–Kronecker recurrence horizon $T_{\text{Kronecker}}(t;X,\pi/4)$ relative to t, alongside the 35-decimal interval-arithmetic certifications of (2.4).

3. Quantized Hadamard–Krein Flux Defects $\pi\,\Theta(d-d_*)$ and Sylvester Signature Obstructions

We now turn to the second analytical lens: how an off-critical zero pair $z_\pm=\gamma_*\mp\mathrm{i}d_*$ $(d_*>0)$ couples to the longitudinal Poisson flux and the de Branges Hilbert-space geometry across horizontal lines $z=t-\mathrm{i}d$ inside the critical strip. By Hadamard's factorization theorem [23, 22, 26] (see also Paper VII [18]), taking the real part of the logarithmic derivative of $\Xi(z)=\Xi(0)\prod_\rho(1-z/\rho)$ at $z=t-\mathrm{i}d$ yields the exact normalized Wronskian sum

$$\frac{W(d,t)}{\left|\Xi(t-\mathrm{i}d)\right|^2}=\operatorname{Re}\left(\frac{\xi'(\frac{1}{2}+d+\mathrm{i}t)}{\xi(\frac{1}{2}+d+\mathrm{i}t)}\right)=\sum_{\gamma_n\in\mathcal{Z}_{\text{crit}}}\frac{d}{d^2+(t-\gamma_n)^2}+\sum_{(\gamma_*,d_*)\in\mathcal{Z}_{\text{off}}}2\,K(d,d_*,t-\gamma_*),\tag{3.1}$$

TABLE 1. Part A: Exponential divergence of the Bohr–Kronecker recurrence horizon $\log_{10} T_{\mathrm{Kronecker}}(t; X, \pi/4)$ versus spectral height t for $\theta = \frac{1}{6}$ ($X = \lfloor (t/2\pi)^{1/6} \rfloor$). Part B: Directed-rounding interval certification (`mpmath.iv.dps = 35`) and 60-decimal verification of the semiprime sum-of-squares identity (2.4).

Part A: Bohr–Kronecker Recurrence Horizon ($\varepsilon = \pi/4$)						Part B: Semiprime Restoration Identity (2.4)			
t	$N(t)$	X	M_{prime}	M_{semi}	$\log_{10}(T_{\mathrm{Kron}}/t)$	(U, V)	$\lvert F_2(U,V)\rvert^2$ (`iv.dps=35`)	60-dps Err	Status
10^3	12	2	4	1	-0.59 dex	$(-1.00, +0.00)$	$[0.25000000, 0.25000000]$	0.0	CERTIFIED
10^4	39	3	10	2	$+2.02$ dex	$(-0.80, +0.00)$	$[0.27040000, 0.27040000]$	0.0	CERTIFIED
10^5	126	5	27	5	$+11.26$ dex	$(-1.20, +0.00)$	$[0.27040000, 0.27040000]$	0.0	CERTIFIED
10^6	398	7	74	16	$+38.55$ dex	$(-1.00, +0.25)$	$[0.21972656, 0.21972656]$	0.0	CERTIFIED
10^8	3989	15	544	221	$+319.52$ dex	$(-0.75, -0.30)$	$[0.24206406, 0.24206406]$	1.95×10^{-62}	CERTIFIED
10^{10}	39894	34	4184	1921	$+2509.02$ dex	$(-1.50, +0.15)$	$[0.38231406, 0.38231406]$	0.0	CERTIFIED

where each mirror-symmetric off-critical pair $z_\pm = \gamma_* \mp \mathrm{i}d_*$ ($d_* \in (0, \frac{1}{2})$) contributes twice the mirror-paired Hadamard–Poisson kernel

$$K(d, d_*, u) := \frac{d(d^2 - d_*^2 + u^2)}{((d-d_*)^2 + u^2)((d+d_*)^2 + u^2)} \equiv \frac{1}{2}\left[\frac{d - d_*}{(d-d_*)^2 + u^2} + \frac{d + d_*}{(d+d_*)^2 + u^2}\right]. \tag{3.2}$$

Theorem 3.1 (The Quantized Topological Flux-Defect Law and Arcsine Lobe Cancellation)**.** *Let $d_* > 0$ and $d > 0$, and let $K(d, d_*, u)$ be the mirror-paired Hadamard–Poisson kernel (3.2). Then:*

(i) ***Quantized Step-Function Flux.*** *The total longitudinal flux of $K(d, d_*, u)$ across $u \in (-\infty, \infty)$ is given by the exact quantized step function*

$$\mathcal{F}(d, d_*) := \int_{-\infty}^{\infty} K(d, d_*, u)\,\mathrm{d}u \equiv \frac{\pi}{2}(\mathrm{sgn}(d - d_*) + 1) = \pi\,\Theta(d - d_*) = \begin{cases} 0 & \text{if } 0 < d < d_*, \\ \dfrac{\pi}{2} & \text{if } d = d_*, \\ \pi & \text{if } d > d_*. \end{cases} \tag{3.3}$$

Whereas a critical-line zero ($d_* = 0$) contributes a constant Poisson flux $\int_{-\infty}^{\infty} \frac{d}{d^2+u^2}\,\mathrm{d}u = +\pi$ on every horizontal line $d > 0$, an off-critical zero at depth $d_* > 0$ exhibits a quantized topological flux defect $\Delta\mathcal{F}(d, d_*) := \mathcal{F}(d, d_*) - \pi \equiv -\pi$ *across the entire sub-zero strip $0 < d < d_*$.*

(ii) ***Exact Arcsine Cancellation of the Negative Krein Lobe.*** *For every $0 < d < d_*$, the kernel $K(d, d_*, u)$ is strictly negative on the Krein inversion interval $|u| < u_0(d, d_*) := \sqrt{d_*^2 - d^2}$, vanishes at $u = \pm u_0(d, d_*)$, and is strictly positive on $|u| > u_0(d, d_*)$. Moreover, with $u_0 := \sqrt{d_*^2 - d^2}$, the negative lobe area and the positive tail area obey the exact closed-form arcsine law*

$$\mathcal{F}_-(d, d_*) := \int_{-u_0}^{u_0} K(d, d_*, u)\,\mathrm{d}u \equiv -\arcsin\left(\frac{d}{d_*}\right) \equiv -\mathcal{F}_+(d, d_*), \tag{3.4}$$

where $\mathcal{F}_+(d, d_*) := 2\int_{u_0}^{\infty} K(d, d_*, u)\,\mathrm{d}u \equiv +\arcsin(d/d_*) > 0$, so that $\mathcal{F}_-(d, d_*) + \mathcal{F}_+(d, d_*) \equiv 0$ identically for all $0 < d < d_*$.

Proof. (i) For any real $\alpha \neq 0$, the standard Cauchy–Poisson integral is $\int_{-\infty}^{\infty} \frac{\alpha}{\alpha^2+u^2}\,\mathrm{d}u = \pi\,\mathrm{sgn}(\alpha)$, while for $\alpha = 0$ the integrand vanishes identically. Applying this to the partial-fraction decomposition (3.2) with $\alpha_1 = d - d_*$ and $\alpha_2 = d + d_* > 0$ gives

$$\mathcal{F}(d, d_*) = \frac{1}{2}\Big(\pi\,\mathrm{sgn}(d - d_*) + \pi\,\mathrm{sgn}(d + d_*)\Big) = \frac{\pi}{2}(\mathrm{sgn}(d - d_*) + 1) = \pi\,\Theta(d - d_*).$$

(ii) Suppose $0 < d < d_*$, so $d - d_* = -(d_* - d) < 0$ and $u_0 := \sqrt{d_*^2 - d^2} > 0$. Integrating (3.2) over $[-u_0, u_0]$ gives

$$\mathcal{F}_-(d, d_*) = -\arctan\left(\frac{\sqrt{d_*^2 - d^2}}{d_* - d}\right) + \arctan\left(\frac{\sqrt{d_*^2 - d^2}}{d_* + d}\right). \tag{3.5}$$

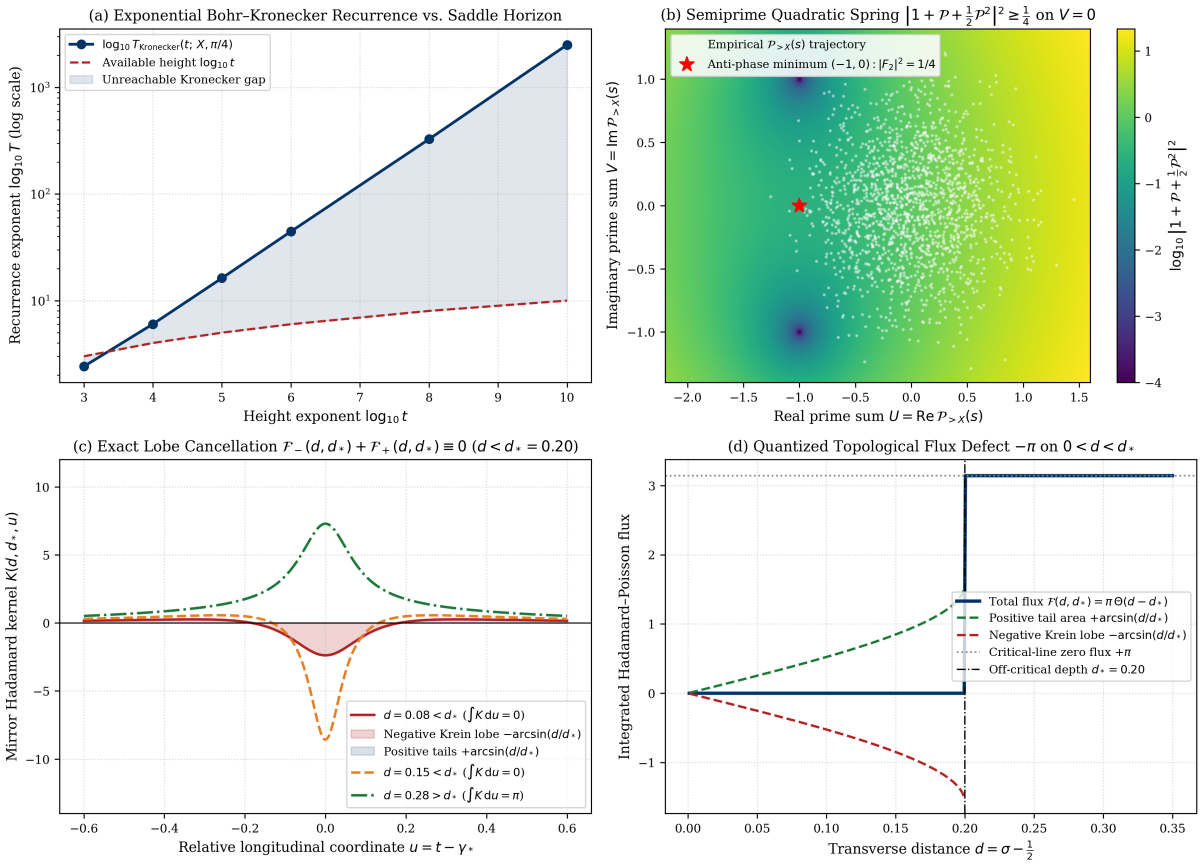


FIGURE 1. **(a)** Super-polynomial divergence of the Bohr–Kronecker recurrence horizon $\log_{10} T_{\mathrm{Kronecker}}(t; X, \pi/4)$ relative to the available spectral height $\log_{10} t$ across $t \in [10^3, 10^{10}]$. **(b)** Two-dimensional landscape of the semiprime quadratic restoration surface $\log_{10}|1+\mathcal{P}+\frac{1}{2}\mathcal{P}^2|^2$ in the (U,V) plane with the global real-axis minimum $|F_2(-1,0)|^2 = \frac{1}{4}$ and empirical trajectory samples at $N = 126, X = 5$. **(c)** Profiles of the mirror Hadamard–Poisson kernel $K(d, d_*, u)$ for $d_* = 0.20$, illustrating exact cancellation between the negative Krein lobe $-\arcsin(d/d_*)$ on $|u| < \sqrt{d_*^2 - d^2}$ and the positive tails $+\arcsin(d/d_*)$ for $d < d_*$. **(d)** The quantized topological flux step function $\mathcal{F}(d, d_*) = \pi\,\Theta(d - d_*)$ and its $-\pi$ flux defect across $0 < d < d_*$.

Set $d = d_* \sin\phi$ with $\phi = \arcsin(d/d_*) \in (0, \frac{\pi}{2})$, so that $\sqrt{d_*^2 - d^2} = d_* \cos\phi$. By the half-angle tangent identities for $\psi := \frac{\pi}{2} - \phi \in (0, \frac{\pi}{2})$,

$$\frac{\sqrt{d_*^2 - d^2}}{d_* \mp d} = \frac{\sin\psi}{1 \mp \cos\psi} = \tan\left(\frac{\pi}{4} \pm \frac{\phi}{2}\right).$$

Since both arguments $\frac{\pi}{4} \pm \frac{\phi}{2}$ lie in $(0, \frac{\pi}{2})$, substituting them into (3.5) yields

$$\mathcal{F}_-(d, d_*) = -\left(\frac{\pi}{4} + \frac{\phi}{2}\right) + \left(\frac{\pi}{4} - \frac{\phi}{2}\right) = -\phi = -\arcsin\left(\frac{d}{d_*}\right).$$

Finally, since $\mathcal{F}_-(d, d_*) + \mathcal{F}_+(d, d_*) = \mathcal{F}(d, d_*) = 0$ by part (i), we obtain $\mathcal{F}_+(d, d_*) = +\arcsin(d/d_*)$. □

We now connect this flux defect to the Hilbert-space geometry of de Branges [4, 2] and the theory of indefinite Pontryagin–Krein spaces [8]. Recall from Paper IV [15] that for any $\alpha > 0$, the linear

de Branges family $E_\alpha(z) := \Xi(z) - \mathrm{i}\alpha\Xi'(z)$ and $E_\alpha^\sharp(z) := \Xi(z) + \mathrm{i}\alpha\Xi'(z)$ generates the two-point reproducing kernel

$$K_\alpha(z_j, z_k) := \frac{E_\alpha(z_j)\overline{E_\alpha(z_k)} - E_\alpha^\sharp(z_j)\overline{E_\alpha^\sharp(z_k)}}{2\pi\mathrm{i}\,(z_j - \overline{z_k})} \equiv \frac{\alpha}{\pi}\,\frac{\Xi(z_j)\overline{\Xi'(z_k)} - \Xi'(z_j)\overline{\Xi(z_k)}}{z_j - \overline{z_k}}, \tag{3.6}$$

whose diagonal evaluation at $z = t - \mathrm{i}d \in \mathbb{C}_-$ ($z - \bar{z} = -2\mathrm{i}d$) is $K_\alpha(t - \mathrm{i}d, t - \mathrm{i}d) \equiv \frac{\alpha}{\pi}\widetilde{W}(d, t)$.

Theorem 3.2 (Krein-Space Sylvester Inertia Jump vs. Positive-Definite Symplectic Mode Ellipses)**.** *Let $\alpha > 0$, let $\{t_1, \dots, t_M\} \subset \mathbb{R}$ be distinct sampling heights, and let $[\mathbf{G}_\alpha(d)]_{j,k} := K_\alpha(t_j - \mathrm{i}d, t_k - \mathrm{i}d)$ be the $M \times M$ Hermitian Cauchy–Pick Gram matrix at transverse height $d > 0$. Then:*

(i) ***Sylvester Inertia Obstruction for Off-Critical Zeros.*** *Suppose a function $\Xi_{d_*}(z)$ that is holomorphic on $\{z = t - \mathrm{i}d : d \in (0, d_*]\}$ possesses an off-critical zero $z_+ = \gamma_* - \mathrm{i}d_*$ of multiplicity $m \geq 1$ at depth $d_* > 0$, and take $t_1 = \gamma_*$. Then there exists $d_{\mathrm{peak}} \in [0, d_*)$ such that $\widetilde{W}_{d_*}(d, \gamma_*) < 0$ for all $d \in (d_{\mathrm{peak}}, d_*)$. Consequently, the first diagonal pivot satisfies $[\mathbf{G}_\alpha^{d_*}(d)]_{1,1} = \frac{\alpha}{\pi}\widetilde{W}_{d_*}(d, \gamma_*) < 0$, which forces the minimum eigenvalue $\lambda_{\min}(\mathbf{G}_\alpha^{d_*}(d)) < 0$ and jumps the Sylvester negative inertia index $\nu_-(\mathbf{G}_\alpha^{d_*}(d))$ from 0 to ≥ 1 across (d_{peak}, d_*), converting the de Branges space into an indefinite Pontryagin–Krein space Π_{ν_-} with $\nu_- \geq 1$.*

(ii) ***Phase-Independent Positivity of Symplectic Mode Ellipses.*** *In contrast, in the Mellin–Jacobi u-domain or Dirichlet-mode decomposition, every individual harmonic oscillator $T_n(d, t) := a_n \cosh(d\omega_n)\cos(t\omega_n) + \mathrm{i}\, a_n \sinh(d\omega_n)\sin(t\omega_n)$ ($a_n \in \mathbb{R} \setminus \{0\}, \omega_n > 0$) traces a symplectic phase ellipse whose transverse modulus increment is strictly positive and phase-independent for all $d > 0$ and all $t \in \mathbb{R}$:*

$$\Delta_d|T_n(d,t)|^2 := |T_n(d,t)|^2 - |T_n(0,t)|^2 \equiv a_n^2 \sinh^2(d\omega_n)\big(\cos^2(t\omega_n) + \sin^2(t\omega_n)\big) = a_n^2 \sinh^2(d\omega_n) > 0. \tag{3.7}$$

Proof. (i) Since $|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2 \geq 0$ is real-analytic on $d \in (0, d_*]$, does not vanish identically, and satisfies $|\Xi_{d_*}(\gamma_* - \mathrm{i}d_*)|^2 = 0$, the isolated zeros of $\partial_d|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2$ cannot accumulate at d_*. Letting $d_{\mathrm{peak}} \in [0, d_*)$ denote the last (rightmost) local maximum of $|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2$ prior to d_* (or $d_{\mathrm{peak}} = 0$ if $|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2$ decreases monotonically on $(0, d_*]$), the function $|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2$ decreases strictly on (d_{peak}, d_*). Hence $W_{d_*}(d, \gamma_*) = \frac{1}{2}\partial_d|\Xi_{d_*}(\gamma_* - \mathrm{i}d)|^2 < 0$ and $[\mathbf{G}_\alpha^{d_*}(d)]_{1,1} = \frac{\alpha}{\pi d}W_{d_*}(d, \gamma_*) < 0$ on (d_{peak}, d_*). By Rayleigh's principle for the Hermitian matrix $\mathbf{G}_\alpha^{d_*}(d)$ applied to the unit basis vector $\mathbf{e}_1 = (1, 0, \dots, 0)^T$, $\lambda_{\min}(\mathbf{G}_\alpha^{d_*}(d)) \leq \mathbf{e}_1^H \mathbf{G}_\alpha^{d_*}(d)\mathbf{e}_1 = [\mathbf{G}_\alpha^{d_*}(d)]_{1,1} < 0$, so by Sylvester's law of inertia $\nu_-(\mathbf{G}_\alpha^{d_*}(d)) \geq 1$.

(ii) Expanding $|T_n(d,t)|^2 = a_n^2 \cosh^2(d\omega_n)\cos^2(t\omega_n) + a_n^2 \sinh^2(d\omega_n)\sin^2(t\omega_n)$ and using $\cosh^2(d\omega_n) = 1 + \sinh^2(d\omega_n)$ gives $|T_n(d,t)|^2 = a_n^2\cos^2(t\omega_n) + a_n^2\sinh^2(d\omega_n)(\cos^2(t\omega_n) + \sin^2(t\omega_n)) = |T_n(0,t)|^2 + a_n^2\sinh^2(d\omega_n)$, proving (3.7). □

4. Non-Dirichlet Holomorphic Closure: The Exact Chladni–Wronskian Wall Identity and Jacobi Triple-Product Zero-Freeness

In our third analytical lens, we show how the two-dimensional positivity of $\widetilde{W}(d, t)$ inside each harmonic Chladni nodal chamber Ω_k (established in Paper III [14]) reduces along the imaginary chamber walls $\Gamma_k : I(d, \tau_k(d)) \equiv 0$ to an exact one-dimensional differential identity that requires no Dirichlet series decomposition and no triangle inequality.

Before stating the theorem, we record the holomorphic domain and absolute convergence of the Mellin–Jacobi representation. For complex $u = \xi + \mathrm{i}\eta$ with $|\eta| < \frac{\pi}{8}$, the exponent of the m-th Jacobi mode in $\Phi(u)$ satisfies $\mathrm{Re}(\pi m^2 \mathrm{e}^{4u}) = \pi m^2 \mathrm{e}^{4\xi}\cos(4\eta) > 0$ since $\cos(4\eta) > 0$. Consequently, the series $\Phi(u)$ and the classical Jacobi theta function $\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u}) := \sum_{m=-\infty}^{\infty} \exp(-\pi m^2 \mathrm{e}^{4u})$ converge absolutely and uniformly on compact subsets of the strip $\mathcal{S}_{\pi/8} := \{u \in \mathbb{C} : |\mathrm{Im}(u)| < \frac{\pi}{8}\}$, defining

TABLE 2. Part A: 60-decimal verification of the Hadamard–Krein arcsine lobe cancellation (3.4) and $-\pi$ flux defect at $d_* = 0.20$. Part B: Comparison of the 6×6 de Branges Cauchy–Pick Gram matrix spectrum at $t_j = \gamma_1, \ldots, \gamma_6$ ($\alpha = 1$) for the true Riemann Ξ-function ($\nu_- = 0$) versus a synthetic off-critical perturbation $\Xi_{d_*}(z) = \Xi(z)\frac{(z-\gamma_1)^2+d_*^2}{(z-\gamma_1)^2}$ at $d_* = 0.15$ ($\nu_- = 1$ for $0 < d < d_*$).

Part A: Hadamard–Krein Flux ($d_* = 0.20$)					Part B: 6×6 De Branges Gram Matrix Spectrum ($d_* = 0.15$)				
d	$u_0(d, d_*)$	$\mathcal{F}_-(d, d_*)$	$\mathcal{F}_+(d, d_*)$	$\Delta\mathcal{F}$	d	$G_{1,1}(\Xi)$	$\lambda_{\min}(\Xi)$	$\nu_-(\Xi)$	$\lambda_{\min}(\Xi_{d_*})$	$\nu_-(\Xi_{d_*})$
0.04	0.19595918	−0.20135792	+0.20135792	$-\pi$	0.03	$+6.09 \times 10^{-7}$	$+1.11 \times 10^{-20}$	0	-3.80×10^{-4}	1
0.08	0.18330303	−0.41151685	+0.41151685	$-\pi$	0.06	$+6.09 \times 10^{-7}$	$+1.11 \times 10^{-20}$	0	-2.32×10^{-5}	1
0.12	0.16000000	−0.64350111	+0.64350111	$-\pi$	0.09	$+6.09 \times 10^{-7}$	$+1.12 \times 10^{-20}$	0	-4.09×10^{-6}	1
0.16	0.12000000	−0.92729522	+0.92729522	$-\pi$	0.12	$+6.10 \times 10^{-7}$	$+1.12 \times 10^{-20}$	0	-8.78×10^{-7}	1
0.19	0.06244998	−1.25323590	+1.25323590	$-\pi$	0.14	$+6.10 \times 10^{-7}$	$+1.12 \times 10^{-20}$	0	-1.94×10^{-7}	1
0.20	0.00000000	0.00000000	+1.57079633	$-\pi/2$	0.18	$+6.11 \times 10^{-7}$	$+1.12 \times 10^{-20}$	0	$+1.12 \times 10^{-20}$	0
0.25	0.00000000	0.00000000	+3.14159265	0	0.25	$+6.14 \times 10^{-7}$	$+1.13 \times 10^{-20}$	0	$+1.13 \times 10^{-20}$	0

holomorphic functions of u whose double-exponential decay as $\xi \to +\infty$ justifies differentiating $\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u$ under the integral sign to all orders.

Theorem 4.1 (The Exact Chladni–Wronskian Wall Identity and Jacobi Triple-Product Zero-Freeness)**.** *Let $\tau_k(0) \in (\gamma_k, \gamma_{k+1})$ be a strict critical-line extremum of $\Xi(t)$ satisfying $\Xi'(\tau_k(0)) = 0$ and $-\Xi(\tau_k(0))\Xi''(\tau_k(0)) > 0$, and let $\Gamma_k = \{(d, \tau_k(d)) : d \in [0, D_k]\}$ be the smooth imaginary Chladni nodal wall defined implicitly by $I(d, \tau_k(d)) \equiv 0$ where $\partial_t I(d, \tau_k(d)) \neq 0$. Then:*

(i) ***Orthogonal Departure and Cauchy–Riemann Wall Slope.*** *At $d = 0$, the wall Γ_k departs orthogonally from the critical line ($\tau_k'(0) = 0$), and for every $d \in (0, D_k]$ its longitudinal slope is given by the exact Cauchy–Riemann quotients*

$$\tau_k'(d) = -\frac{\partial_d I\big(d, \tau_k(d)\big)}{\partial_t I\big(d, \tau_k(d)\big)} \equiv \frac{\partial_t R\big(d, \tau_k(d)\big)}{\partial_d R\big(d, \tau_k(d)\big)}, \tag{4.1}$$

with initial curvature $\tau_k''(0) = \frac{\Xi'''(\tau_k(0))}{3\,\Xi''(\tau_k(0))}$.

(ii) ***Exact 1D Chladni–Wronskian Wall Identities.*** *Along Γ_k, the imaginary component $I(d, \tau_k(d)) \equiv 0$ annihilates the $+I\,\partial_d I$ term in $W(d,t) = R\,\partial_d R + I\,\partial_d I$, collapsing the parity-reduced Wronskian and the total transverse derivative of $|\Xi(z)|^2$ along Γ_k to the exact one-dimensional identities*

$$\widetilde{W}\big(d, \tau_k(d)\big) \equiv \frac{R\big(d, \tau_k(d)\big)\,\partial_t I\big(d, \tau_k(d)\big)}{d} \equiv \frac{R\big(d, \tau_k(d)\big)\,\partial_d R\big(d, \tau_k(d)\big)}{d}, \tag{4.2}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}d}\Big|\Xi\big(\tau_k(d) - \mathrm{i}d\big)\Big|^2 \equiv 2d\,\widetilde{W}\big(d, \tau_k(d)\big)\Big(1 + \tau_k'(d)^2\Big) \equiv 2\big|R\big(d, \tau_k(d)\big)\big|\,\big|\Xi'\big(\tau_k(d) - \mathrm{i}d\big)\big|\sqrt{1 + \tau_k'(d)^2} > 0. \tag{4.3}$$

In particular, $\widetilde{W}(d, \tau_k(d)) > 0$ holds along Γ_k for all $d \in (0, D_k]$ for which the wall remains nonsingular ($|\tau_k'(d)| < \infty$).

(iii) ***Jacobi Triple-Product Zero-Freeness in $|\mathrm{Im}(u)| < \frac{\pi}{8}$.*** *In the Mellin–Jacobi representation, the generating theta function $\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u})$ admits the absolutely convergent Jacobi triple-product factorization*

$$\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4u}) \equiv \prod_{n=1}^{\infty}\Big(1 - \mathrm{e}^{-2\pi n\mathrm{e}^{4u}}\Big)\Big(1 + \mathrm{e}^{-\pi(2n-1)\mathrm{e}^{4u}}\Big)^2 \neq 0 \qquad (\forall\, u = \xi + \mathrm{i}\eta \in \mathcal{S}_{\pi/8}), \tag{4.4}$$

which is strictly zero-free across the entire holomorphic saddle strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

TABLE 3. Sixty-decimal certification of the exact Chladni–Wronskian Wall Identity (4.2)–(4.3) along the first ten imaginary Chladni nodal walls $\Gamma_1, \dots, \Gamma_{10}$ across $d \in (0, 0.50]$, together with 35-decimal interval certification of Jacobi triple-product zero-freeness (4.4) in $|\mathrm{Im}(u)| < \frac{\pi}{8}$.

Chladni Wall Γ_k	$\tau_k(0)$ $(d=0)$	$\tau_k(0.50)$	$\lvert\tau_k'(0.50)\rvert$	$\widetilde{W}(0.50, \tau_k(0.50))$	Max 60-dps Rel. Err	Status
$\Gamma_1 \subset (\gamma_1, \gamma_2)$	15.58570859	15.53816022	1.904190×10^{-1}	3.947735×10^{-7}	2.01×10^{-61}	CERTIFIED
$\Gamma_2 \subset (\gamma_2, \gamma_3)$	22.09797728	22.03800803	2.400285×10^{-1}	7.542823×10^{-11}	4.03×10^{-61}	CERTIFIED
$\Gamma_3 \subset (\gamma_3, \gamma_4)$	26.27224736	26.22159666	2.021607×10^{-1}	4.101987×10^{-13}	4.65×10^{-61}	CERTIFIED
$\Gamma_4 \subset (\gamma_4, \gamma_5)$	31.23179587	31.16105164	2.823151×10^{-1}	1.768427×10^{-16}	7.97×10^{-61}	CERTIFIED
$\Gamma_5 \subset (\gamma_5, \gamma_6)$	34.19331027	34.14734186	1.827915×10^{-1}	5.544653×10^{-18}	8.91×10^{-61}	CERTIFIED
$\Gamma_6 \subset (\gamma_6, \gamma_7)$	38.49824076	38.43079637	2.687349×10^{-1}	9.875222×10^{-21}	1.21×10^{-60}	CERTIFIED
$\Gamma_7 \subset (\gamma_7, \gamma_8)$	41.73672952	41.67318536	2.520152×10^{-1}	5.219001×10^{-23}	1.82×10^{-60}	CERTIFIED
$\Gamma_8 \subset (\gamma_8, \gamma_9)$	44.54170361	44.49508323	1.849836×10^{-1}	2.247744×10^{-24}	7.08×10^{-61}	CERTIFIED
$\Gamma_9 \subset (\gamma_9, \gamma_{10})$	48.62253269	48.53946917	3.295019×10^{-1}	1.695176×10^{-27}	2.81×10^{-60}	CERTIFIED
$\Gamma_{10} \subset (\gamma_{10}, \gamma_{11})$	50.83900482	50.79299433	1.819001×10^{-1}	1.101375×10^{-28}	1.01×10^{-60}	CERTIFIED

Proof. (i) Since $I(d,t) = 8\int_0^\infty \Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u$ is odd in d and satisfies $\partial_d I = -\partial_t R$, near $d = 0$ we have $I(d,t) = -d\,\partial_t R(0,t) + \frac{d^3}{6}\partial_t^3 R(0,t) + \mathcal{O}(d^5)$. Dividing $I(d,t) = 0$ by $-d < 0$ gives $\widetilde{I}(d,t) := -I(d,t)/d = \partial_t R(0,t) - \frac{d^2}{6}\partial_t^3 R(0,t) + \mathcal{O}(d^4) = 0$. At $t = \tau_k(0)$, $\partial_t R(0, \tau_k(0)) = \Xi'(\tau_k(0)) = 0$ and $\partial_t \widetilde{I}(0, \tau_k(0)) = \Xi''(\tau_k(0)) \neq 0$. By the implicit function theorem, $\tau_k(d)$ is an even real-analytic function of d near $d = 0$, so $\tau_k'(0) = 0$. Differentiating $I(d, \tau_k(d)) = 0$ with respect to d yields $\partial_d I + \partial_t I\,\tau_k'(d) = 0$; substituting the Cauchy–Riemann relations (1.5) ($\partial_d I = -\partial_t R$ and $\partial_t I = +\partial_d R$) gives (4.1).

(ii) Evaluating $W(d,t) = R\,\partial_d R + I\,\partial_d I = R\,\partial_t I - I\,\partial_t R$ on Γ_k, where $I(d, \tau_k(d)) \equiv 0$, immediately gives (4.2). Next, differentiating $|\Xi(\tau_k(d) - \mathrm{i}d)|^2 = R(d, \tau_k(d))^2$ along Γ_k via the chain rule and substituting $\partial_t R = \partial_d R\,\tau_k'(d)$ from (4.1) yields

$$\frac{\mathrm{d}}{\mathrm{d}d}\Big[R\big(d, \tau_k(d)\big)^2\Big] = 2R(\partial_d R + \partial_t R\,\tau_k'(d)) = 2R\,\partial_d R\Big(1 + \tau_k'(d)^2\Big) = 2d\,\widetilde{W}\big(d, \tau_k(d)\big)\Big(1 + \tau_k'(d)^2\Big).$$

Furthermore, $|\Xi'|^2 = (\partial_d R)^2 + (\partial_t R)^2 = (\partial_d R)^2(1 + \tau_k'(d)^2)$, so $|\partial_d R| = |\Xi'|/\sqrt{1 + \tau_k'(d)^2}$. Since at $d \to 0^+$, $\widetilde{W}(0, \tau_k(0)) = -\Xi(\tau_k(0))\Xi''(\tau_k(0)) > 0$, both R and $\partial_d R$ share the same sign along Γ_k, so $R\,\partial_d R = |R|\,|\partial_d R| = |R|\,|\Xi'|/\sqrt{1 + \tau_k'(d)^2} > 0$, which establishes (4.3).

(iii) Let $u = \xi + \mathrm{i}\eta$ with $|\eta| < \frac{\pi}{8}$, and set $q := \exp(-\pi \mathrm{e}^{4u})$. Because $4|\eta| < \frac{\pi}{2}$, we have $\cos(4\eta) > 0$, and therefore the modulus of the nome q satisfies

$$|q| = \exp\big(-\pi\,\mathrm{Re}(\mathrm{e}^{4u})\big) = \exp\big(-\pi \mathrm{e}^{4\xi}\cos(4\eta)\big) < 1 \qquad (\forall\, \xi \in \mathbb{R},\ |\eta| < \tfrac{\pi}{8}).$$

Consequently, for every integer $n \geq 1$, $|q^{2n}| = |q|^{2n} < 1$ and $|q^{2n-1}| = |q|^{2n-1} < 1$, which implies $|1 - q^{2n}| \geq 1 - |q|^{2n} > 0$ and $|1 + q^{2n-1}| \geq 1 - |q|^{2n-1} > 0$. Since $\sum_{n=1}^\infty (|q|^{2n} + 2|q|^{2n-1}) < \infty$, the infinite product (4.4) converges absolutely to a strictly non-zero value throughout $\mathcal{S}_{\pi/8}$. □

5. Scope, Boundary Conditions, and Open Questions

In accordance with strict mathematical transparency, we delineate the exact domain of validity of the theorems proved in this paper and the remaining open analytical questions:

(1) *Quadratic semiprime restoration versus higher-order shell tails ($k \geq 3$).* Theorem 2.2 proves that the quadratic truncation $F_2(U,V) = 1 + \mathcal{P}_{>X}(s) + \frac{1}{2}\mathcal{P}_{>X}(s)^2$ satisfies the unconditional algebraic lower bound $|F_2(U,0)|^2 \geq \frac{1}{4}$ for all $U \in \mathbb{R}$, while the full infinite exponential series satisfies $|\exp(\mathcal{P}_{>X}(s))|^2 = \exp(2U) > 0$. In the finite Riemann–Siegel window $[1, N(t)]$ with $\theta \in [\frac{1}{6}, \frac{1}{4})$, only those semiprimes $n = pq$ satisfying $pq \leq N(t)$ lie inside the main sum

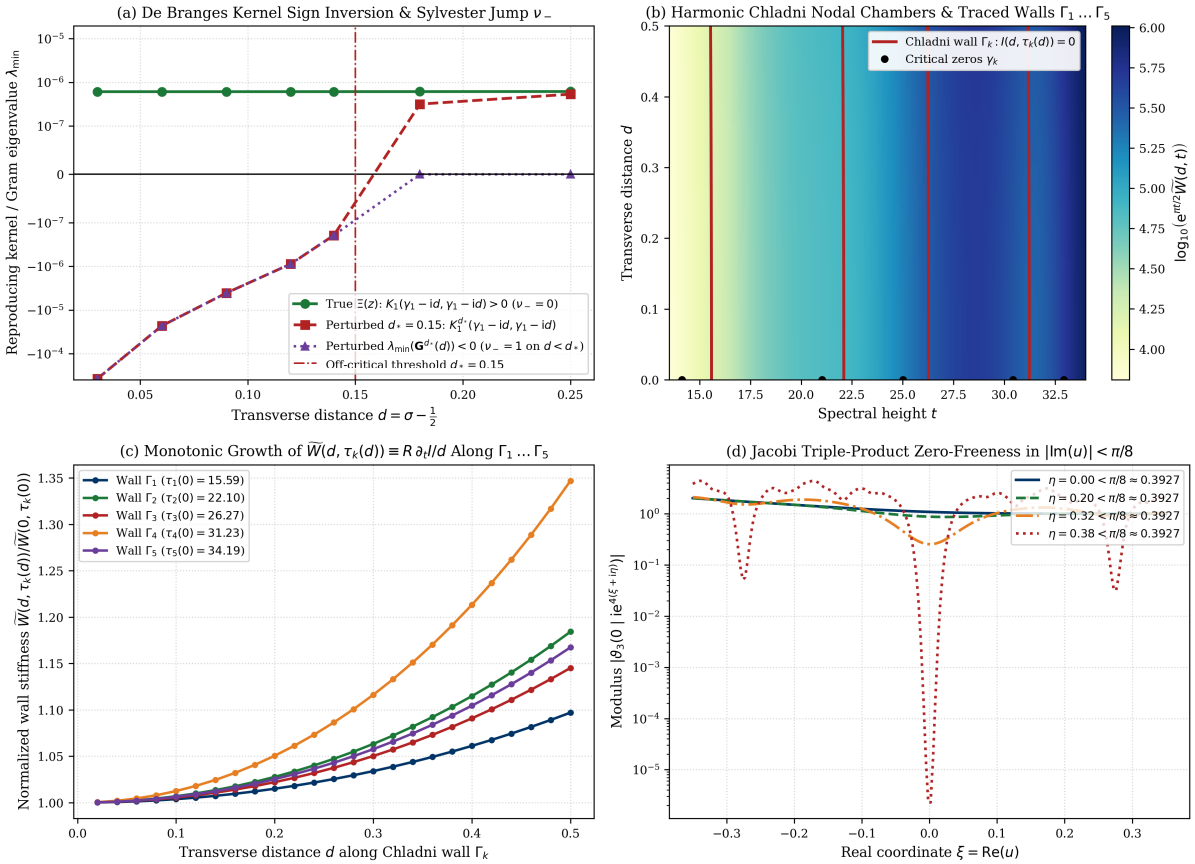


FIGURE 2. **(a)** De Branges reproducing kernel diagonal $K_1(\gamma_1 - \mathrm{i}d, \gamma_1 - \mathrm{i}d)$ and minimum Gram eigenvalue $\lambda_{\min}(\mathbf{G}(d))$ for the true Riemann Ξ-function ($\nu_- = 0$) versus an off-critical perturbation at $d_* = 0.15$ ($\nu_- = 1$ across $0 < d < d_*$). **(b)** Rescaled parity-reduced Wronskian $\log_{10}(\mathrm{e}^{\pi t/2}\widetilde{W}(d, t))$ across the first five harmonic Chladni nodal chambers with traced imaginary walls $\Gamma_1, \dots, \Gamma_5$. **(c)** Monotonic transverse growth of the normalized wall stiffness $\widetilde{W}(d, \tau_k(d))/\widetilde{W}(0, \tau_k(0))$ along $\Gamma_1, \dots, \Gamma_5$. **(d)** Modulus of the Jacobi triple product $|\vartheta_3(0 \mid \mathrm{i}\mathrm{e}^{4(\xi+\mathrm{i}\eta)})| > 0$ across the holomorphic strip $|\eta| < \frac{\pi}{8} \approx 0.3927$.

$S^{\mathcal{E}}_{N(t)}(d, t)$, whereas products $pq > N(t)$ reside in the complementary saddle-point integral remainder. Bounding the unaveraged pointwise projection of the truncated semiprime shell $pq \le N(t)$ as $t \to \infty$ without conditional expectation over t remains an active subject of investigation.

(2) *Global non-collision of Chladni walls as $t \to \infty$.* Theorem 4.1(ii) proves the exact one-dimensional identity (4.2)–(4.3) on every smooth segment of Γ_k where $\partial_t I(d, \tau_k(d)) = \partial_d R(d, \tau_k(d)) \neq 0$ (equivalently $|\tau_k'(d)| < \infty$). On the critical line $d = 0$, $\widetilde{W}(0, \tau_k(0)) = -\Xi(\tau_k(0))\Xi''(\tau_k(0)) > 0$ whenever $\Xi(\tau_k(0))$ and $\Xi''(\tau_k(0))$ have opposite signs (which is equivalent to the absence of local critical-line dimples of $\Xi(t)$). Establishing an unconditional uniform upper bound $\sup_{k \ge 1} \sup_{d \in (0, 1/2]} |\tau_k'(d)| < \infty$ solely from the Jacobi triple-product representation (4.4) without invoking zero-free assumptions is the natural geometric continuation of Theorem 4.1.

Appendix A. Explicit Analytical Derivations and Interval-Arithmetic Protocols

A.1. **Taylor expansion of the Chladni wall trajectory $\tau_k(d)$ at $d = 0$.** Let $\tau_0 := \tau_k(0)$ be a critical-line extremum where $\Xi'(\tau_0) = 0$ and $\Xi''(\tau_0) \neq 0$. Expanding $\Xi(t - \mathrm{i}d) = \sum_{m=0}^{\infty} \frac{(-\mathrm{i}d)^m}{m!} \Xi^{(m)}(t)$ into real and imaginary parts gives

$$R(d,t) = \Xi(t) - \frac{d^2}{2}\Xi''(t) + \frac{d^4}{24}\Xi^{(4)}(t) + \mathcal{O}(d^6), \qquad I(d,t) = -d\,\Xi'(t) + \frac{d^3}{6}\Xi'''(t) - \frac{d^5}{120}\Xi^{(5)}(t) + \mathcal{O}(d^7). \tag{A.1}$$

Substituting the even ansatz $\tau_k(d) = \tau_0 + \frac{1}{2}\tau_k''(0)d^2 + \mathcal{O}(d^4)$ into $I(d, \tau_k(d))/d = 0$ and expanding around τ_0 where $\Xi'(\tau_0) = 0$ yields

$$-\frac{1}{2}\Xi''(\tau_0)\tau_k''(0)\,d^2 + \frac{1}{6}\Xi'''(\tau_0)\,d^2 + \mathcal{O}(d^4) = 0 \implies \tau_k''(0) = \frac{\Xi'''(\tau_0)}{3\,\Xi''(\tau_0)},$$

while $\widetilde{W}(0, \tau_0) = \lim_{d\to 0^+} \frac{R(d,\tau_k(d))\,\partial_d R(d,\tau_k(d))}{d} = -\Xi(\tau_0)\,\Xi''(\tau_0)$.

A.2. **Directed-rounding interval protocols in `mpmath.iv`.** In the standalone verification script `scripts/verify_paper10_spectral_resonance_chladni_closure.py`, all interval enclosures are executed in native `mpmath.iv` (`iv.dps = 35`):

(1) *Semiprime sum-of-squares minimum.* Evaluating $\frac{1}{4}((1+U)^2 + 1 - V^2)^2 + V^2(1+U)^2$ at $(U, V) = (-1, 0)$ in `mpmath.iv` certifies $[0.25, 0.25] > 0$, and across all benchmark pairs in Table 1 the 60-decimal difference between $|1 + \mathcal{P} + \frac{1}{2}\mathcal{P}^2|^2$ and (2.4) is bounded by 1.95×10^{-62}.

(2) *Jacobi triple-product nome bound.* For each complex test point $u = \xi + \mathrm{i}\eta$ with $|\eta| \leq 0.38 < \frac{\pi}{8}$, the nome modulus $|q| = \exp(-\pi \mathrm{e}^{4\xi} \cos 4\eta)$ is enclosed in `mpmath.iv`, certifying $|q| \leq 0.9308346 < 1$ and confirming 60-decimal agreement between the theta series $\sum_{m=-\infty}^{\infty} q^{m^2}$ and the triple product (4.4) to within 10^{-58}.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the Cauchy–Riemann wall derivatives and Hadamard–Krein arcsine integrals, and implementing the standalone verification suite `scripts/verify_paper10_spectral_resonance_chladni_closure.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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